

Table 5: Comparison of classification accuracy (%) between the different variants of the proposed approach and FGS method (FGS-orig) for different values of ϵ

Type	0	0.02	0.04	0.06	0.08	0.1
Baseline	89.4	67.5	49.6	41.2	37.3	34.7
FGS-orig (?)	88.7	86.4	84.1	81.4	80.5	77.1
FGS-inter	90.9	87.79	83.85	79.65	74.69	69.92
Ours-orig	91.2	87.95	83.84	79.11	73.66	68.37
Ours-joint	91.5	86.07	81.38	75.72	70.25	64.76

the strength of adversarial FGS perturbation added to the input image. From these results, we make the following observations: (1) Approaches based on perturbing intermediate layers (FGS-inter, Ours-orig, Ours-joint) improve the performance on the original data significantly as compared to perturbing only the input but they marginally decrease the adversarial test performance. (2) On the other hand, perturbing only the input layer (FGS-orig) yields the best adversarial test performance among the compared approaches while performing marginally worse than the baseline on the original test data. These observations indicate the possibility of a trade-off that exists between adversarial robustness and regularization effect over clean data. Referring to the toy example described earlier, the singular value analysis performed there also supports our claim that methods which impart adversarial robustness tend to suppress sensitivity of the model in all the dominant directions; while the proposed approach provides a nice trade-off by retaining those directions which are essential for modeling the variations in the training data. This ensures that our approach results in better regularization performance on clean data while providing comparable robustness on adversarial data.

Results on WideResNet models: Wide Residual Networks are recently proposed deep architectures that generated state of the art results on CIFAR-10 and CIFAR-100 datasets. In this experiment, we use their publicly available implementation and train them from randomly initialized weights using the proposed approach using the parameter settings described in the experiments section. Specifically, the Ours-joint approach described in the previous section is used for training. As data augmentation, we applied flipping and random cropping as done in their native implementation. The results are shown in Table 6. For the compared methods, we quote their published accuracy values. It can be observed that the proposed approach results in improved performance compared to both the baseline model and the model regularized with dropout. This demonstrates the generalization ability of our approach to state of art deep models.

Table 6: Classification error rates (%) on CIFAR-10 and CIFAR-100 for WideResNet (WRN) architectures. Our results are reported as average of 5 runs. For comparison we provide the published WRN baseline results. (*) denotes the results obtained by a single run.

Model	#params	CIFAR-10	CIFAR-100
WRN-28-10 (?)	36.5M	4.00	19.25
WRN-28-10 with dropout (?)	36.5M	3.89	18.85
WRN-40-10 with dropout* (?)	51.0M	3.8	18.3
WRN-28-10 - Ours	36.5M	3.62 $\pm$ 0.05	17.1 $\pm$ 0.1

Generalization to non-image signals: In this work, we have considered adversarial perturbations in the space of natural images. The existence of adversarial perturbations has been shown to exist in other types of signals that occur in speech recognition ((?), (?)) and language processing tasks (?). While the focus of the this paper has been images, there exists a natural extension of our framework to the above modalities. Such an extension is trivially possible since end-to-end learning systems such as deep networks are used in speech and language tasks as well. As future work, we propose to extend the current approach to explore robustness aspects of deep networks trained on modalities other than images.

Summary and Conclusion

While the behavior of CNNs to adversarial data has generated some intrigue in computer vision since the work of (?), its effects on deeper networks have not been explored well. We observe that adversarial perturbations for hidden layer activations generalize across different samples and we leverage this observation to devise an efficient regularization approach that could be used to train very deep architectures. Through our experiments and analysis we make the following observations: (1) Contrary to recent methods which are inconclusive about the role of perturbing intermediate layers of a DNN in adversarial training, we have shown that for very deep networks, they play a significant role in providing a strong regularization (2) The aggregated adversarial effect of perturbing intermediate layer activations is much stronger than perturbing only the input (3) Significant improvement in classification accuracy entails capturing more variations in the data distribution while adversarial robustness can be improved by suppressing the unnecessary variations learned by the network. By providing an efficient adversarial training approach that could be used to regularize very deep models, we hope that this can inspire more robust network designs in the future.

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